USN | | | | | | | | | | | [22CSBS219]

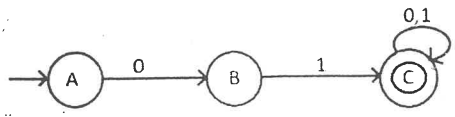
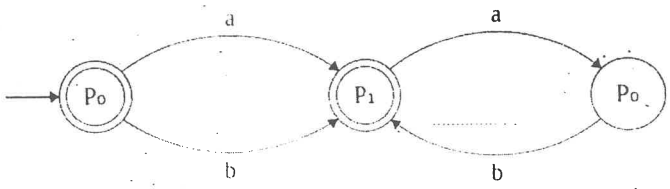
THIRD SEMESTER B.Tech EXAMINATION, DECEMBER -2023

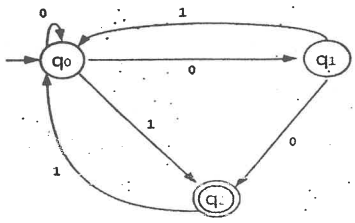
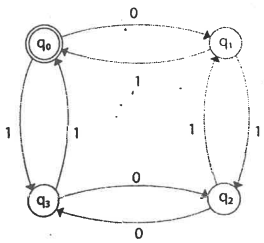
FORMAL LANGUAGE AND AUTOMATA THEORY

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 X 2=10**		
1.	**a.**	Define alphabet, string, powers of an alphabet and concatenation of strings in FSA	L1	CO1	**02**
	b.	Draw the Transition table of a given Finite state Automata 	L3	CO2	**02**
	c.	State the pumping lemma for CFLs	L1	CO3	**02**
	d.	Write the Formal definition of Turing Machine	L6	CO4	**02**
	e.	Define NP Complete	L1	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain the difference between DFA and NFA	L2	CO1	**05**
		OR			
	b.	Draw the Transition table of a given NFA transition diagram and Explain it. 	L3	CO1	**05**
3.	**a.**	Draw DFA accepting all strings over w ∈ {a,b}* which contains "aba" as a sub string	L3	CO1	**05**
		OR			
	b.	Construct NFA for the given Regular Expression 01(0+1)	L3	CO2	**05**
4.	**a.**	Construct the CFG L(G) = $\{0^n1^{n+1} \mid n \geq 0\}$. the grammar having the following production rules: S → 0S1 |1|ε	L3	CO2	**05**
		OR			
	b.	Show that the following grammars are ambiguous. E → E+E|E*E|(E)|id	L5	CO2	**05**

5.	**a.**	Define Push Down Automata (PDA) Explain it	L3	CO3	**05**
		OR			
	b.	Design a PDA for accepting a language $\{0^n \mid n>=1\}$.	L3	CO3	**05**
6.	**a.**	Explain the basic model of Turing machine	L2	CO4	**05**
		OR			
	b.	Explain Turing Machine variant Two Stack PDA	L2	CO4	**05**
7.	**a.**	State and prove Church-Turing Thesis	L5	CO5	**05**
		OR			
	b.	Prove that universal language L_u is a recursively enumerable language but not recursive.	L5	CO5	**05**
		PART-C **06 X 10=60**			
8.	**a.**	Construct an equivalent DFA for the following NFA 	L3	CO1	**10**
		OR			
	b.	Find the Languages accepted by the given NFA as in Fig 	L1	CO1	**10**
9.	**a.**	Convert Regular Grammar into Finite automata S→aX | bX A→aX | bX| bY B→aY | bY | a | b	L3	CO2	**10**
		OR			
	b.	Construct NFA for the given Regular Expression $(1+0)(0 + 10)^*$	L3	CO2	**10**

10.	**a.**	Find the given grammar G is ambiguous or not. **i.** $S \rightarrow aSb \mid SS$ and $S \rightarrow \varepsilon$ **ii.** $A \rightarrow AA$, $A \rightarrow (A)$ and $A \rightarrow a$	L1	CO2	**10**
		OR			
	b.	Convert the given CFG to CNF. Consider the given grammar G1: $S \rightarrow a \mid aA \mid B$ $A \rightarrow aBB \mid \varepsilon$ $B \rightarrow Aa \mid b$	L4	CO2	**10**
11.	**a.**	Construct pushdown automata for the givenlanguages. Acceptance either by empty stack or by final state. $\{ a^n b^m c^m \mid m, n \in N \}$	L3	CO3	**10**
		OR			
	b.	Convert Context Free Grammar into Push Down automata $E \rightarrow E+E$ $E \rightarrow E*E$ $E \rightarrow a$	L3	CO3	**10**
12.	**a.**	Construct a TM that accepts input starts with 'aaab' where all input made of the combination of {a,b}	L3	CO4	**10**
		OR			
	b.	Describe the following **i.** Recursive language **ii.** Recursively enumerable language	L2	CO4	**10**
13.	**a.**	Explain the following problems **i.** NP Hard **ii.** NP Complete	L2	CO5	**10**
		OR			
	b.	Consider the correspondence system as given below A = (b, bab^3, ba) and B = (b^3, ba, a). The input set is $\Sigma = \{0, 1\}$. Find the solution.	L1	CO5	**10**